

We observe that the graph G (applicable to questions 1–3) has the following vertices:

$$V(G) = \{a, b, c, d, e, f\},$$

and edges

$$ab, bc, ad, be, cf, de, ef, bd, bf.$$

1. Find the chromatic polynomial $P(G, \lambda)$.

We wish to choose a color for b : there are λ choices.

Trivially (and by def), every neighbor of b must avoid that color. The neighbors of b are

$$a, c, d, e, f.$$

After coloring b , the remaining edges among these five vertices are

$$ad, \quad de, \quad ef, \quad fc.$$

So these five vertices form a path

$$a - d - e - f - c.$$

Thus, once b is colored, we are coloring a path on 5 vertices using only the remaining $\lambda - 1$ colors.

A path on 5 vertices has chromatic polynomial

$$x(x - 1)^4.$$

Substituting $x = \lambda - 1$, we get

$$(\lambda - 1)(\lambda - 2)^4.$$

Therefore,

$$P(G, \lambda) = \lambda(\lambda - 1)(\lambda - 2)^4.$$

So the chromatic polynomial is

$$\boxed{P(G, \lambda) = \lambda(\lambda - 1)(\lambda - 2)^4.}$$

2. In how many ways can G be colored with (at most) 5 colors?

The number of proper colorings with colors chosen from a set of 5 labeled colors is

$$P(G, 5).$$

Using the polynomial from part (1), we find:

$$P(G, 5) = 5(5 - 1)(5 - 2)^4 = 5 \cdot 4 \cdot 3^4.$$

Since

$$3^4 = 81,$$

we get

$$P(G, 5) = 5 \cdot 4 \cdot 81 = 1620.$$

Finally, we find the number of proper colorings with at most 5 available colors is 1620.

3. Find the chromatic number $\chi(G)$.

The graph contains a triangle (technically, it can be argued it's entirely composed of triangles, but this is sufficient):

$$a - b - d.$$

Therefore at least 3 colors are necessary:

$$\chi(G) \geq 3.$$

Now, we will evaluate the chromatic polynomial at $\lambda = 3$:

$$P(G, 3) = 3(3 - 1)(3 - 2)^4 = 3 \cdot 2 \cdot 1^4 = 6 > 0.$$

Since $P(G, 3) > 0$, the graph is properly 3-colorable. Therefore,

$$\chi(G) \leq 3.$$

Combining both inequalities,

$$\chi(G) = 3.$$

So,

$$\boxed{\chi(G) = 3.}$$

4. Suppose

$$f(\lambda) = \lambda^6 - 11\lambda^5 + 45\lambda^4 - 85\lambda^3 + 74\lambda^2 - 24\lambda$$

is the chromatic polynomial of a graph F . Decide whether F could be planar.

From lecture, we know to test planarity, use the Four Color Theorem: every planar graph is properly colorable with at most 4 colors. Therefore, if F were planar, we would have

$$f(4) > 0.$$

Compute:

$$f(4) = 4^6 - 11 \cdot 4^5 + 45 \cdot 4^4 - 85 \cdot 4^3 + 74 \cdot 4^2 - 24 \cdot 4.$$

Now

$$4^2 = 16, \quad 4^3 = 64, \quad 4^4 = 256, \quad 4^5 = 1024, \quad 4^6 = 4096,$$

so

$$f(4) = 4096 - 11(1024) + 45(256) - 85(64) + 74(16) - 96.$$

That is,

$$f(4) = 4096 - 11264 + 11520 - 5440 + 1184 - 96 = 0.$$

Since

$$f(4) = 0,$$

the graph F has no proper 4-coloring. But every planar graph must have a proper 4-coloring. Therefore F cannot be planar.

So the conclusion is F cannot be planar.

(Trivially, we find that one can factor

$$f(\lambda) = \lambda(\lambda - 1)^2(\lambda - 2)(\lambda - 3)(\lambda - 4),$$

so F would require 5 colors.)